DATA ITEM DESCRIPTION

Form Approved
OMB No. 0704-0188

1. TITLE	2 IDENTIFICATION NUMBER
NUCLEAR SURVIVABILITY MAINTENANCE/ SURVEILLANCE PLAN	DI-NUOR-81025

3. DESCRIPTION / PURPOSE

3.1 The purpose of the Nuclear Survivability Maintenance/Surveillance Plan is as follows: (1) Survivability Maintenance - to describe the special activities and precautions needed to assure the system's nuclear survivability is not degraded and (2) Surveillance - to describe the inspection and test procedures necessary to verify that the system's nuclear survivability is not degraded.

4 APPROVAL DATE (YYMMDD)	5. OFFICE OF PRIMARY RESPONSIBILITY (OPR)	6a DTIC APPLICABLE	6b. GIDEP APPLICAB
900829	A/SLCHD-NW-TS		

7. APPLICATION / INTERRELATIONSHIP

7.1 This Data Item Description (DID) contains the format and content preparation instructions for the data product generated by the specific and discrete task requirements as delineated in the contract.

7.2 This DID is applicable to contracts which contain a requirement for nuclear survivability in the Full-Scale Engineering Development acquisition phase or in the Production acquisition phase.

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8. APPROVAL LIMITATION	9a. APPLICABLE FORMS	9b. AMSC NUMBER
		A4981

10. PREPARATION INSTRUCTIONS

10.1 General. The Nuclear Survivability Maintenance/Surveillance Plan shall describe the operational procedures, design features, materials, and piece parts that control the system's nuclear survivability; shall delineate how these shall be preserved during operation, maintenance, repair, and product improvement of the system; and shall delineate how the system's nuclear survivability can be periodically monitored during deployment. The Nuclear Survivability Maintenance/Surveillance Plan shall reflect the philosophy of the Nuclear Survivability Program Plan (DI-NUOR-80156A) and shall be based on data from the Nuclear Survivability Design Parameters Report (NUOR-80927) and the Nuclear Survivability Assurance Plan (NUOR-80926).

10.2 Format. The Nuclear Survivability Maintenance/Surveillance Plan format shall be contractor selected. Unless effective presentation would be degraded, the initially used format arrangement shall be used for all subsequent submissions.

10.3 Content. The content of the Nuclear Survivability Maintenance/Surveillance Plan shall be as follows:

10.3.1 Degradation Prevention Techniques. Identify the techniques used to prevent degradation of the nuclear survivability of the system in normal operation, maintenance, repair or product improvement, to include the technologies, designs, piece parts selected for the system based on nuclear survivability considerations and the rationale for these choices.

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11. DISTRIBUTION STATEMENT

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

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APPLICATION/INTERRELATIONSHIP (Continued)

3 This DID shall be used in conjunction with DI-NUOR-80156 and DI-NUOR-80926.

.4 The delivery schedule for the Nuclear Survivability Maintenance/Surveillance Plan should be consistent with the availability of engineering drawings, parts lists, and other technical data and information for use in the analysis of maintenance tasks under the Logistic Support Analysis (LSA) process. Allowing time for the preparation of the Nuclear Survivability Program Plan (DI-NUOR-80156A) and the Nuclear Survivability Design Parameters Report (DI-NUOR-80927), delivery of the Nuclear Survivability Maintenance/Surveillance Plan should not be later than 90-180 days after contract award depending on the complexity of the system and total program length. The final report should be submitted 60 days prior to the end of the full scale development (FSD) phase or development/proveout phase.

7.5 Portions of the Nuclear Survivability Maintenance/Surveillance Plan are used in the preparation of training/technical manuals and other logistic support documentation (reference MIL-STD-1388-1 and MIL-STD-1388-2). Therefore, the solicitation should delineate and the contract should require incorporation of data from the Nuclear Survivability Maintenance/Surveillance Plan into the final version of the training/technical manuals and other logistic support documentation.

7.6 This DID supersedes DI-R-1763A.

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10. PREPARATION INSTRUCTIONS (Continued)

10.3.2 Exclusion Considerations. Identify the specific considerations used to exclude certain designs, piece parts or piece part technologies, components, or materials, including the reason for these exclusions.

10.3.3 Critical Aspects. Identify the most critical aspects of the Nuclear Survivability Maintenance/Surveillance Plan, the consequences of the alternative approaches considered, and the risks associated with the recommended approach.

10.3.4 Operational Considerations. The nuclear survivability operational considerations shall be identified, including:

10.3.4.1 The maintenance and operating procedures which must be performed prior to a nuclear event to insure that if a nuclear event occurs an acceptable nuclear response of the system may be anticipated. These procedures shall include deployment guidelines, protective procedures (such as orientation of the equipment, configuration of the various items of the system), protective devices (such as canvas and shelters), camouflage, and the readiness availability requirements for the repairable/replaceable parts, modules, or components which are vulnerable to the nuclear event.

10.3.4.2 The maintenance and operating procedures which must be performed after a nuclear event (e.g., cycling the system power, replacing fuzes or other components, resetting circuit breakers and reloading logic data) to insure operational restoration of the system.

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10. PREPARATION INSTRUCTIONS (Continued)

10.3.5 Hardware Considerations. Identification of the assemblies, subassemblies, circuits, modules, components, piece parts, materials, and processes that are critical to the system's survival in a nuclear environment, the specific parameter values or qualities to be maintained for these items, and the methods for maintaining them.

10.3.6 Surveillance Procedures. The nuclear survivability surveillance procedures shall include:

10.3.6.1 A listing of materiel (e.g., piece parts, materials, components, circuits, structures, subassemblies, assemblies, and systems) that may undergo hardness degradation as a result of the conditions of operational life along with the associated degradation modes of concern. Identify the estimated probability of occurrence (or priority listing) of the operational life hardness degradation modes postulated and the impact on system mission should those degradations actually occur.

10.3.6.2 A description of the inspection and test procedures for the materiel. Provide detailed information concerning tests and inspections to be performed; monitor points and rationale for choices; ambient conditions; modes of test sample operation and sample configuration during the test with rationale; sample size, the number and kinds of Government furnished equipment requested; the interface requirements or the interaction of the test item with the rest of the system; and the data to be recorded. Provide the acceptability criteria for the tests, and remedial actions where the acceptability criteria are not met. Provide the methodology for interpretation of test results to an assessment of how the system responds to the specified environments. Identify the facilities to be employed, special instrumentation, measurement techniques, calibration procedures, test methods and data analysis, and specifics of the surveillance schedule to be followed.

10.3.6.3 The specifics of the surveillance schedule shall include the time after deployment before the initiation of the various actions, the frequency of testing and inspection, the reporting schedule, the number of personnel and skills required, and the timing of this surveillance schedule with respect to the normal maintenance schedule. Address the possibilities for incorporating/coordinating the surveillance tests and inspections with routine maintenance/repair activities.

10.3.6.4 A description of the proposed data management for surveillance information, including identification of appropriate government and, or contractor responsibilities.

10.3.7 Documentation. The Nuclear Survivability Maintenance/Surveillance Plan shall include a flow chart which shows the origin of the information/data (e.g., drawings or documents), what information flows through the logistic support analysis records and where the different types of information or data have been recorded in the equipment documentation. Describe how the methods, procedures and data for the Nuclear Survivability Maintenance/Surveillance Plan have been incorporated into the equipment documentation (i.e., technical manuals, provisioning documents, training plans, support and test equipment plans, etc.).

10.3.7.1 Identify the drawings and documentation which have been marked with the notation HCI - for Hardness Critical Item or HCP - for Hardness Critical Process (reference DOD-STD-100). The complete HCI and HCP listing shall identify the total means by which survivability has been explicitly incorporated within the design of that Configuration Item (CI). The HCI and HCP criteria are as follows:

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PREPARATION INSTRUCTIONS (Continued)

.7.1.1 An HCI is defined to be an item of hardware or software that satisfies or more of the following rationales:

a. Functionally required hardware (i.e., hardware included in system design to accomplish any engineering requirement other than nuclear survivability) whose response to the specified nuclear environments would cause a degradation in system survivability unless additional provisions are included in item specification, design, manufacture, item selection process, etc.

b. Functionally required hardware or software identified or modified to also provide protection for the system or any of its elements against the specified nuclear environments.

c. Hardness dedicated hardware or software included in system design solely to help satisfy the specified survivability requirements.

d. Hardware items (at the level of application) to which an HCP is applied.

e. A subassembly or higher level of assembly which contains one or more HCIs.

10.3.7.1.2 A HCP is defined as a process, procedure, or specification which is critical to the system nuclear survivability, and which, if changed, could satisfy normal operational requirements, and yet degrade system survivability.

10.3.8 Configuration Control Actions. The final version of the Nuclear Survivability Maintenance/Surveillance Plan shall incorporate all changes from Configuration Control Board actions which impact the system's nuclear survivability.

10.3.9 Survivability Maintenance Data. Identify all drawings and documents that have been or will be annotated in support of nuclear survivability maintenance.

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